

Name of the Student _____

Scholar Number _____

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
END TERM EXAMINATION-APRIL 2024

Course: B. Tech

Semester: IV

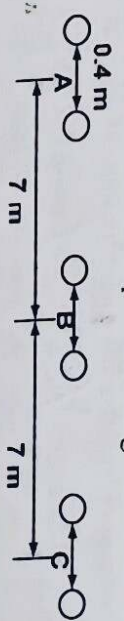
Subject Name: POWER SYSTEM

Subject Code: EE-221

Time Duration: 3 Hours

Maximum Marks: 50

Subject Coordinators: Dr. Yogendra Kumar and Dr. More Raju

S.No	Question	Marks	CO
1	A) Find the inductive reactance of a 3-phase bundled conductor line with 2-conductors per phase with spacing of 40 cm. phase to phase separation is 7m in horizontal configuration. All conductors are ACSR with diameter of 3.5 cm. Compare the above value with that of an equivalent single conductor line.  B) A two-conductor, single-phase line operates at 50 Hz. The diameter of each conductor is 2 cm and are spaced at 3 m apart. Then calculate - Capacitance of each conductor to neutral per km, - Line to line capacitance - Capacitive susceptance to neutral per km.	7	CO-1
2	A) What is string efficiency? Why is it necessary to have high string efficiency? How can it be achieved? B) A three phase transmission line is supported by a string insulator of 3 units. The disc near to power conductor has a voltage of 16 kV and the voltage across second disc is 12 kV. Then find - Ratio of mutual to ground capacitance - System line voltage - String efficiency	5	CO-2
3	A) A 30 km long three phase overhead line delivers 6 MW at 11 kV at 0.8 lagging pf. The line loss is 10% of power delivered. Line inductance is 1.2 mH per km per phase. Find sending end voltage and voltage regulation. B) Derive A, B, C, D constants of medium length transmission line (for both nominal-T and nominal-π methods) and hence prove $AD - BC = 1$. C) A three phase, 50 Hz, overhead transmission line delivers 20 MW at 0.8 pf lagging at 66 kV. The resistance and inductive reactance of the transmission line per phase are 10Ω and 20Ω while capacitive admittance is 4×10^{-4} S. Calculate	6	CO-3

Name of the Student

Rishu

Scholar Number

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	(i) Sending end voltage (ii) Sending end current and power factor (iii) Transmission efficiency Note: Use nominal T-method.		
	A) What is Kelvin's law. Explain it with graphical representation as well state it's limitations. B) A DC two line distributor AB, 600 m long is fed at 440 V from substation A and at 430 V from substation B, the loads are: 100 A at C, 150 m from A 200 A at D, 150 m from C 250 A at E, 50 m from D 300 A at F, 100 m from E. If each line conductor has a resistance of 0.01Ω per 100 m, calculate the current supplied from substations A and B and the voltage across each load.	5	CO-4
4			
	A) A 3-core, 3-phase metal sheathed cable had (i) capacitance of $1 \mu\text{F}$ between shorted conductor and sheath; (ii) capacitance between two shorted conductors with the sheath and the third conductor $0.6 \mu\text{F}$. Find the capacitance between any two conductors. Also find the capacitance between any two shorted conductors and the third conductor. B) Give the graphical representation of symbols required for the various types of apparatus and elements of a substation. Draw a layout of 33 kV substation.	3	CO-4
5		4	CO-5